NYC Health + Hospitals

OBGYN — AI Scheduling Platform

Functional Requirements & Demonstration Guide

FOR DEMONSTRATION ONLY — NOT FOR CLINICAL USE — SYNTHETIC DATA

This guide pairs each capability of the demo with its **functional requirement** (what the system must do) and a **how-to** for demonstrating it live, ending with the **expected outcome**. Requirements trace to the OBGYN AI Scheduling use cases (UC1–UC8) and business rules (BR-1…BR-14).

All data is synthetic. No PHI is used. Names are fictional; phones use the reserved 555-01xx range; MRNs are SYN-xxxx. Every AI recommendation is human-approved and audited.

Application UI	`https://nychhc-frontend-iis-ai-ui.apps.ocp419.crucible.iisl.com`
Backend API	`https://nychhc-copilot-iis-ai-ai.apps.ocp419.crucible.iisl.com`
Dashboards	`Grafana — https://grafana-iis-ai-ui.apps.ocp419.crucible.iisl.com (admin / Demo1234#)`
Platform	OpenShift AI (ocp419.crucible.iisl.com) · CPU-only · synthetic data

AI solution by IIS · iistech.com

FR-1 · Role-based access (UC: DR-01)

> **FUNCTIONAL REQUIREMENT**
>
> The system shall present role-tailored views for Scheduler, Approver, Provider, and Leadership, and shall restrict actions to those permitted for the active role (BR-9).

How to demonstrate

1. Open the Application UI; accept the self-signed-certificate warning.
2. In the left sidebar under **Role & context**, switch between **Scheduler** (persona Selamawit), **Approver**, **Provider**, and **Leadership**.
3. Observe the tab bar and persona change with each role.

> **✓ EXPECTED OUTCOME**
>
> **Scheduler** → Dashboard, Schedule, No-Show Risk, Approvals, Assistant.
>
> **Approver** → Dashboard, PTO, Approvals, Assistant.
>
> **Provider** → Schedule, PTO, Assistant (read-only for writes).
>
> **Leadership** → Reporting, Dashboard, Assistant. The synthetic-data banner shows on every screen.

FR-2 · Predict patient no-show risk (UC1)

> **FUNCTIONAL REQUIREMENT**
>
> The system shall compute a no-show probability per appointment and classify it red / amber / green, with human-readable explanatory factors (BR-2).
>
> Risk thresholds shall be configurable without code (BR-3). If the model is unavailable the system shall degrade gracefully and never fabricate scores.

How to demonstrate

1. As **Scheduler**, open the **No-Show Risk** tab.
2. Filter by All / Red / Amber / Green, or search by patient, MRN, or provider.
3. Read the **Top factors** and recommended action for each flagged appointment.

> **✓ EXPECTED OUTCOME**
>
> A risk-scored list for today: RED > 65%, AMBER 35–65%, GREEN < 35%.
>
> Example: **Mei Chen** 75% RED — “3 prior no-shows, New OB, no text on file” → Call patient.
>
> If the No-Show model endpoint is down, a **degraded-mode** banner appears and rules-based scores are shown (clearly labelled, not model output).

FR-3 · View & manage the schedule (UC: DR-02/03)

> **FUNCTIONAL REQUIREMENT**
>
> The system shall let an authorised user view provider calendars and create, modify, or cancel appointments, with validation that prevents double-booking.

How to demonstrate

1. As **Scheduler**, open the **Schedule** tab and click **New appointment**.
2. Walk the drawer: specialty (Obstetrics) → provider (Dr. Amara Okonkwo) → an open slot → confirm.
3. Use **Modify** or **Cancel** and search by patient name or MRN to change an appointment.

> ✓ **EXPECTED OUTCOME**
>
> Slots show as Open / Booked / Blocked, computed from the live schedule; double-booking is rejected.
>
> Cancelling frees the slot and suggests higher-risk patients to re-offer it to.

FR-4 · PTO impact & coverage conflict (UC4)

> **FUNCTIONAL REQUIREMENT**
>
> The system shall show the downstream appointment impact of a leave request before approval, and shall surface coverage conflicts where concurrent leave drops a service line below its configured minimum (BR-4, BR-6).
>
> The system shall not silently approve a request that leaves a service window uncovered.

How to demonstrate

1. As **Approver**, open the **PTO** tab and select the pending request for **Dr. Amara Okonkwo** (Jun 16–20).
2. Review the impacted appointments and the auto-resolvable vs. manual split.
3. Read the coverage-conflict banner and the recommended mitigation.

> ✓ **EXPECTED OUTCOME**
>
> 4 appointments impacted; 3 auto-resolvable (reassign to a peer obstetric provider).
>
> **Coverage conflict**: Inpatient OB needs 2 on service but falls short on 3 days — overlaps with Dr. Rachel Stein's leave. Mitigation: stagger leave, pull a peer onto service, or deny one.
>
> Approving an uncovered window is **blocked** until explicitly overridden.

FR-5 · Human-in-the-loop approval & audit (UC6)

> **FUNCTIONAL REQUIREMENT**

> The system shall require an explicit human decision before any AI-proposed action executes — nothing auto-executes (BR-1).
>
> Every approved or rejected action shall be recorded and attributable to a named user with a timestamp (BR-10).

How to demonstrate

1. Trigger a recommendation — on the **PTO** tab click **Apply all auto**, or ask the assistant to apply reassignments (FR-6).
2. Open the **Approvals** tab (Scheduler or Approver).
3. In **Pending approvals**, click **Approve** or **Reject**; then read the **Audit trail**.

> ✓ **EXPECTED OUTCOME**
>
> On Approve, the action executes and an audit row is written: action, decision, outcome, the deciding **user** and a **timestamp** (e.g. *pto_reassign · approved · executed · chat:Scheduler*).
>
> On Reject, nothing executes and the rejection is logged. A Provider cannot approve writes.

FR-6 · Natural-language scheduling assistant (UC5)

> **FUNCTIONAL REQUIREMENT**
>
> The system shall answer plain-language scheduling questions grounded in real data (BR-8), maintain conversation context across turns, clarify ambiguous requests, and decline out-of-scope or clinical/PHI requests.
>
> The assistant shall only perform actions permitted for the user's role (BR-9), routing any change through the approval gate (FR-5).

How to demonstrate

1. Open the **Assistant** tab; use a suggested chip or type a question.
2. Run the three headline asks, then the follow-up, in the same conversation.

```
Which OB providers have openings?
What's the no-show rate by provider?
Put Dr. Okonkwo on PTO 6/16-6/20 and show the impact
apply all auto (follow-up — uses the previous answer)
```

Then test the guardrails:

1. Ask *“who can cover Dr. Okonkwo?”* (no date) → expect a clarifying question.
2. Ask *“show me the nursing schedule”* → expect a polite out-of-scope decline.
3. Use **Clear** to reset the conversation (also clears server-side context).

> ✓ **EXPECTED OUTCOME**

Openings → Obstetrics providers + soonest slots (Stein 06-09 08:00, Okonkwo 08:20, Rahman 09:00).

No-show rate → average risk per provider with the high-risk (red) count.

PTO impact → same impact + coverage conflict as the PTO tab.

“apply all auto” → uses the prior turn and applies the reassignments via the approval gate: “Done — approved and applied 3 reassignment(s)… recorded in the audit trail.”

Ambiguous asks are clarified; out-of-scope / PHI asks are declined.

FR-7 · Epic data access via the MCP adapter (UC8)

FUNCTIONAL REQUIREMENT

The AI layer shall access scheduling data only through a tool-based adapter and shall never hold Epic credentials or query Epic directly (BR-12).

The adapter shall return FHIR-shaped data so a real Epic FHIR endpoint can replace the synthetic source with no caller changes (BR-14); failures shall return typed errors, not fabricated data.

How to demonstrate

1. List the callable tools (any HTTP client; -k for the self-signed cert):

```
curl -k <BACKEND>/api/mcp/tools
```

2. Invoke one tool and inspect the FHIR-shaped result:

```
curl -k -X POST <BACKEND>/api/mcp/call \
-H 'content-type: application/json' \
-d '{"tool":"get_provider_schedule","args":{"provider_id":"p1","date":"2026-06-09"}}'
```

✓ EXPECTED OUTCOME

/api/mcp/tools lists get_patient_appointments, get_provider_schedule, check_slot_availability, get_risk_scores, check_pto_impact.

/api/mcp/call returns FHIR R4 Appointment/Slot JSON; an unknown tool or unreachable source returns a typed error with *degraded: true* (no fabricated data).

FR-8 · Leadership reporting (department-level)

FUNCTIONAL REQUIREMENT

The system shall present department-level performance — throughput, coverage reliability, and capacity utilisation — to support the Chair / CCO business case (no transactional detail required).

How to demonstrate

1. Switch to the **Leadership** role and open the **Reporting** tab.
2. For operational charts open **Grafana** → dashboard **NYCHHC — Workforce & Patient-Flow** (sign in admin / Demo1234#).

> ✓ **EXPECTED OUTCOME**
>
> Reporting tiles: coverage reliability, at-risk appointments, open coverage conflicts, capacity utilisation.
>
> Grafana: no-show risk mix, no-show rate by provider, the PTO queue, and the roster.

FR-9 · Resilience, safety & reset

> **FUNCTIONAL REQUIREMENT**
>
> The system shall remain operable if a model endpoint is unavailable (graceful rules fallback) and shall never fabricate model output.
>
> All demo data shall be synthetic with the disclaimer on every page and API response, and the environment shall be reproducible between demos.

How to demonstrate

1. In the Assistant, click **Clear** to reset the conversation and server-side context.
2. To rebuild operational data, an operator runs ./destroy.sh then ./deploy.sh (re-seeds the synthetic OBGYN data).
3. Open-ended chat beyond the headline asks uses Claude via Portkey; the deterministic router still answers the headline asks even with no LLM key.

> ✓ **EXPECTED OUTCOME**
>
> The demo never hard-fails: models down → rules fallback + degraded banner.
>
> Every screen and API envelope carries the synthetic-data disclaimer.
>
> Replace <BACKEND> in the commands with the Backend API URL on the cover.